

# HumanInTheLoopMiddleware

Pause the agent execution for human approval.

**Useful for:**

- High-stakes operations (database writing, financial transaction)
- Compliance workflow
- Long-running conversation where the human feedback guide the agent

**Note:** HumanInTheLoopMiddleware requires a `checkpointer` to maintain state across interruptions.

```
from langchain.agents import create_agent
from langchain.agents.middleware import HumanInTheLoopMiddleware
from langgraph.checkpoint.memory import InMemorySaver

def your_read_email_tool(email_id: str) -> str:
    """Mock function to read an email by its ID."""
    return f"Email content for ID: {email_id}"

def your_send_email_tool(recipient: str, subject: str, body: str) -> str:
    """Mock function to send an email."""
    return f"Email sent to {recipient} with subject '{subject}'"

agent = create_agent(
    model="gpt-5.4",
    tools=[your_read_email_tool, your_send_email_tool],
    checkpointer=InMemorySaver(),
    middleware=[
        HumanInTheLoopMiddleware(
            interrupt_on={
                "your_send_email_tool": {
                    "allowed_decisions": ["approve", "edit", "reject"],
                },
                "your_read_email_tool": False,
            }
        ),
    ],
)
```